

SymFM: A Physics-Informed Foundation Model with Dimensionality-Aware Symbolic Regression for Governing Equation Discovery in High-Dimensional Nonlinear Dynamical Systems

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July 2026

Abstract

Discovering governing equations from observational data in high-dimensional nonlinear dynamical systems remains one of the most fundamental unsolved challenges in computational science. Three distinct failure modes prevent existing methods from operating at scale: physics-informed observer methods such as PINN-Obs have never been evaluated beyond low-dimensional settings, leaving their generalisability to real-world systems unknown; KAN-based symbolic regression degrades combinatorially as edge splines grow with state dimension, making equation recovery intractable beyond moderate N ; and PDE foundation models learn solution operators rather than equation structure, producing no interpretable symbolic output. We introduce **SymFM** (Symbolic Foundation Model), the first framework to simultaneously satisfy all five properties required for scalable autonomous equation discovery: foundation model pretraining, physics-informed training objectives, symbolic interpretable output, high-dimensional applicability, and zero domain-specific architecture design. SymFM integrates four modules: a PINN-Obs state estimator for partial observations, a Transformer-based physics-informed encoder pretrained to represent equation structure across diverse dynamical system families, a learned sparse active subspace projection that reduces effective input dimensionality from N to $d \ll N$, and a hierarchical KAN symbolic regression head with sparsity-promoted closed-form extraction. Evaluated on three benchmark systems of increasing dimensionality (Lorenz-96 at $N \in \{4, 10, 20, 40\}$, 2D Navier-Stokes at $N = 1024$, and a spatially heterogeneous SEIRD model at $N \in \{50, 250, 500\}$), SymFM achieves 100% equation recovery on Lorenz-96 at $N = 4$ and on 2D Navier-Stokes at $N = 1024$ ($\ell_2 = 0.110$), the only method to recover the governing structure at this scale. At $N = 40$ on Lorenz-96, SymFM reduces derivative approximation error by 49% relative to SINDy (0.972 vs. 1.919). On the spatially heterogeneous SEIRD epidemiological benchmark at $N = 500$, the integrated PINN-Obs module achieves state reconstruction RMSE of 0.003529 on unobserved compartments, constituting the first systematic evaluation of physics-informed observers at this dimensionality. These results suggest that differentiable physical constraints encoded directly into foundation model pretraining can substitute for the domain-specific architectural engineering that has previously made autonomous equation discovery inapplicable at scale, opening the door to proteome- and climate-scale equation surveys.

Keywords: physics-informed machine learning, symbolic regression, Kolmogorov-Arnold Networks, governing equation discovery, high-dimensional dynamical systems, foundation models, PINN-Obs, active subspace methods, Lorenz-96, Navier-Stokes.

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1 Introduction

Physical laws govern every domain of natural science, yet deriving those laws directly from observational data remains one of the most challenging open problems in computational science. From Newton’s laws of motion to Maxwell’s equations, the history of physics is a history of governing equations discovered from observation. Today, the frontier of this endeavour lies in systems of high dimensionality and complexity: turbulent fluid flows governed by the Navier-Stokes equations, coupled climate models with millions of interacting degrees of freedom, and biological regulatory networks of thousands of molecular species. In these domains, governing equations either do not yet exist or are known only approximately. A principled computational method capable of autonomously recovering interpretable governing equations from observational data in these settings would represent a step-change in the pace of scientific discovery, accelerating progress across physics, biology, climate science, and engineering.

The equation discovery landscape is currently divided into three families of methods, each of which fails the high-dimensional challenge in a distinct way. Physics-informed neural networks (PINNs) and their observer variants (PINN-Obs) [Raissi et al., 2019] enforce physical consistency through residual losses and achieve strong state estimation from partial observations. However, PINN-Obs produces no symbolic governing equations and has been evaluated exclusively on low-dimensional benchmark systems with state dimension $N \leq 10$. Its behaviour at $N \gg 10$, the regime of practical interest, is entirely uncharacterised. Symbolic regression methods based on sparse identification (SINDy) [Brunton et al., 2016] and Kolmogorov-Arnold Networks (KANs) [Liu et al., 2025] produce closed-form interpretable expressions and have demonstrated impressive results on standard benchmarks. However, a fundamental limitation persists: as state dimension N grows, the number of learnable edge splines in KAN-based approaches grows with it, training time increases substantially, and the post-hoc symbolic extraction pipeline becomes combinatorially intractable [Somvanshi et al., 2024, Bühler and Guillén-Gosálbez, 2025, Howard et al., 2026]. PDE foundation models such as Poseidon [Herde et al., 2024], Walrus [McCabe et al., 2025], and PI-MFM [Zhu et al., 2025] achieve remarkable generalisation across equation families through large-scale Transformer pretraining, but they learn solution operators mapping initial conditions to trajectories. They produce no symbolic output and, critically, neglect physics when target data is sparse, limiting their reliability in low-data regimes [Medvedev et al., 2026].

The gap left by all three families is precisely the combination of properties that practical autonomous equation discovery requires. No existing method simultaneously possesses a foundation model’s generalisation, a PINN’s physical fidelity, a KAN’s symbolic interpretability, and the scalability to operate at high state dimensions without domain-specific architectural redesign. We introduce SymFM to fill this gap.

Summary of contributions. This paper makes four contributions:

1. **Physics-informed foundation model for equation structure (Section 4.3).** We introduce a Transformer-based encoder pretrained with physics-informed residual losses on a diverse corpus of 50,000 synthetic dynamical systems, targeting equation-space representations rather than solution operators. This is the first foundation model pretraining objective designed specifically for symbolic equation discovery rather than PDE solution.
2. **Dimensionality-aware KAN symbolic regression head (Section 4.5).** We introduce a hierarchical KAN architecture incorporating sparse active subspace projection, reducing effective input dimensionality from N to $d \ll N$ and directly addressing the documented

combinatorial scalability failure of existing KAN-based symbolic regression at high state dimension.

3. **First systematic high-dimensional evaluation of PINN-Obs (Section 5.8).** We provide the first systematic benchmark of PINN-Obs state estimation as a component of an end-to-end equation discovery pipeline across state dimensions from $N = 4$ to $N = 500$, filling an explicit gap in the existing literature on physics-informed observers.
4. **Open-source implementation and standardised benchmarks (Appendix B).** We release the full SymFM implementation and all three benchmark systems with standardised evaluation protocols, enabling reproducible comparison for future work. Code: <https://github.com/Eddiegah/SymFM>.

The remainder of this paper is organised as follows. Section 2 surveys related work across all relevant subfields. Section 3 formalises the governing equation discovery problem and states the three assumptions underlying SymFM. Section 4 presents the complete SymFM architecture with all mathematical derivations. Section 5 describes the experimental setup, baselines, and presents results. Section 6 interprets findings and addresses limitations. Section 7 concludes with future directions.

2 Background and Related Work

2.1 Physics-Informed Neural Networks and PINN-Obs

Physics-informed neural networks (PINNs), introduced by Raissi, Perdikaris, and Karniadakis [Raissi et al., 2019], embed physical laws as residual terms in the training objective, enforcing differential equation constraints without requiring labelled solution data. The approach has been applied to forward problems (solving known PDEs), inverse problems (inferring PDE coefficients from data), and state estimation (reconstructing unobserved system states from partial measurements). The PINN-Obs variant extends PINNs to nonlinear observer design, achieving state reconstruction from partial observations through a physics-constrained neural network that ensures the reconstructed trajectory satisfies the assumed governing equation structure. Crucially, all published evaluations of PINN-Obs operate in low-dimensional settings with $N \leq 10$ state variables. Whether the approach remains effective at high state dimensions, where the identifiability of the governing function from partial observations becomes fundamentally harder, has not been characterised in any published work. The present paper provides this evaluation for the first time as an integral component of the SymFM pipeline.

2.2 Symbolic Regression and Sparse Identification

Sparse identification of nonlinear dynamics (SINDy) [Brunton et al., 2016] frames equation discovery as a sparse regression problem over a user-specified library of candidate functions $\Theta(x)$, recovering governing equations by identifying the sparse coefficient vector that best explains observed time derivatives. SINDy is computationally efficient and produces interpretable closed-form expressions, but requires the user to pre-specify the function library and is sensitive to noise in numerical differentiation of state trajectories. Ensemble-SINDy [Fasel et al., 2022] and weak SINDy [Messenger and Bortz, 2021] partially address these limitations.

Kolmogorov-Arnold Networks (KANs) [Liu et al., 2025] offer a learnable alternative to fixed function libraries by replacing node activations in conventional neural networks with trainable uni-

variate spline functions on edges, enabling flexible and interpretable function approximation. KAN-SR [Bühler and Guillén-Gosálbez, 2025] applies KANs to symbolic regression through hierarchical divide-and-conquer decomposition, achieving competitive results on standard benchmarks. SINDy-KANs [Howard et al., 2026] integrates KANs directly into the SINDy framework, combining the structured sparsity of SINDy with the flexibility of learnable activations. Despite these advances, a shared and fundamental limitation persists across all KAN-based methods: as noted in the KAN survey [Somvanshi et al., 2024], scalability challenges arise for very high input dimensions because the number of edge splines grows with dimensionality, and the training and post-hoc simplification pipeline becomes nontrivial for highly nonlinear or high-dimensional problems. High-dimensional systems therefore remain beyond the practical reach of all existing KAN-based symbolic regression approaches. SymFM addresses this directly through active subspace projection before the KAN head.

2.3 Foundation Models for PDEs and Scientific Machine Learning

The emergence of large-scale Transformer architectures has produced a new paradigm in scientific machine learning: foundation models pretrained across diverse families of differential equations that generalise to new problems without retraining. Poseidon [Herde et al., 2024] achieves strong generalisation across 15 PDE families by pretraining a scalable multi-scale operator on a diverse corpus of PDE datasets. Walrus [McCabe et al., 2025] extends this to a broader range of physical systems including fluid dynamics and magnetohydrodynamics. PI-MFM [Zhu et al., 2025] incorporates physics-informed residual losses directly into pretraining and adaptation, addressing the challenge of physically inconsistent predictions under sparse data. These models represent the current state of the art for PDE solution and represent critical baselines for SymFM.

However, all existing PDE foundation models share a critical limitation for the equation discovery task: they learn solution operators mapping initial conditions and boundary conditions to solution trajectories. They produce no symbolic governing equations and are not designed to represent or extract the structure of the underlying differential operator. A fundamentally different pretraining objective is required to target equation structure rather than solution operators. LLM-SR [Shojaee et al., 2025] and the Think Like a Scientist framework [Yang et al., 2026] approach equation discovery through large language models, using iterative program synthesis to search for symbolic expressions consistent with data. These approaches are flexible and leverage the prior knowledge encoded in LLM pretraining corpora, but are not physics-informed, lack convergence guarantees, and are not designed specifically for the high-dimensional dynamical systems setting that motivates SymFM.

2.4 Active Subspace Methods and Dimensionality Reduction

Active subspace methods [Constantine, 2015] provide a principled framework for identifying the low-dimensional subspaces in input space on which a quantity of interest concentrates most of its variation. The active subspace is defined as the dominant eigenspace of the expected outer product of gradients, and has been applied to sensitivity analysis, surrogate modelling, and Bayesian inference in high-dimensional parameter spaces. Recent work has combined dimensionality reduction with symbolic regression for stochastic dynamical systems [Chen et al., 2024], demonstrating that high-dimensional systems often admit sparse representations in suitable low-dimensional subspaces, and that symbolic regression in the projected space produces more interpretable and generalisable equations than in the full space. SymFM incorporates a learned active subspace projection as a differentiable module trained jointly with the foundation model encoder, extending this approach

to the equation discovery setting and enabling the KAN symbolic regression head to operate in a tractable d -dimensional space regardless of the original state dimension N .

2.5 Positioning SymFM Against Prior Work

Table 1 summarises the unique combination of properties that SymFM achieves relative to all prior methods. No existing method checks all five boxes simultaneously, which constitutes the central novelty claim of this work.

Table 1: Comparison of SymFM against all directly relevant prior methods. Each column represents a property required for practical autonomous equation discovery at scale. $\checkmark$ = fully satisfied, $\times$ = not satisfied, $\sim$ = partially satisfied. SymFM is the only method to satisfy all five.

Method	Foundation Model	Physics-Informed	Symbolic Output	High-Dim Tested	No Domain Arch.
PINN-Obs [Raissi et al., 2019]	$\times$	$\checkmark$	$\times$	$\times$	$\times$
SINDy [Brunton et al., 2016]	$\times$	$\times$	$\checkmark$	$\times$	$\times$
KAN/PISF [Liu et al., 2025]	$\times$	$\sim$	$\checkmark$	$\times$	$\times$
SINDy-KANs [Howard et al., 2026]	$\times$	$\times$	$\checkmark$	$\times$	$\times$
Poseidon [Herde et al., 2024]	$\checkmark$	$\times$	$\times$	$\sim$	$\checkmark$
PI-MFM [Zhu et al., 2025]	$\checkmark$	$\checkmark$	$\times$	$\sim$	$\checkmark$
LLM-SR [Shojaee et al., 2025]	$\checkmark$	$\times$	$\checkmark$	$\times$	$\checkmark$
SymFM (this work)	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$	$\checkmark$

3 Problem Formulation

Definition 1 (Governing Equation Discovery Problem). *Let $x(t) \in \mathbb{R}^N$ denote the state of an autonomous nonlinear dynamical system governed by*

$$\dot{x}(t) = f(x(t); \theta), \quad (1)$$

where $f : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is an unknown smooth vector field and $\theta \in \mathbb{R}^p$ are unknown parameters. We observe only a partial noisy measurement process

$$y(t) = Hx(t) + \epsilon(t), \quad \epsilon(t) \sim \mathcal{N}(0, \sigma^2 I_m), \quad (2)$$

where $H \in \mathbb{R}^{m \times N}$ is a known linear observation matrix with $m \leq N$ and $\sigma > 0$ is the noise level. Given a finite collection of noisy measurements $\{y(t_k)\}_{k=1}^T$ at times $0 = t_1 < t_2 < \dots < t_T$, the governing equation discovery problem is to recover a symbolic expression $\hat{f}$ satisfying:

- (i) **Fidelity:** $\|\dot{x}(t) - \hat{f}(x(t))\|_2 \leq \varepsilon$ for all t in the observation interval, for some tolerance $\varepsilon > 0$;
- (ii) **Interpretability:** $\hat{f}$ is expressible as a closed-form mathematical expression in a standard function library; and
- (iii) **Scalability:** recovery is computationally tractable for $N \gg 1$.

Assumption 1 (Autonomy). *The governing vector field f does not depend explicitly on time t , i.e., $f(x(t); \theta) = f(x(t))$ for all t .*

Assumption 2 (Library Sparsity). *There exists a function library $\Theta(x) = [\theta_1(x), \dots, \theta_M(x)]^\top$ of M candidate basis functions such that $f(x)$ is sparse in Θ : $f(x) = \Xi^\top \Theta(x)$ with Ξ having at most $s \ll M$ nonzero rows per output component.*

Assumption 3 (Active Subspace Existence). *There exists a projection matrix $W \in \mathbb{R}^{d \times N}$ with $d \ll N$ and $WW^\top = I_d$ such that*

$$\|f(x) - f(W^\top Wx)\|_2 \leq \delta \quad (3)$$

for all x in the data manifold, where $\delta \ll \|f(x)\|_2$ is a small approximation error. We call the column space of $W^\top$ the active subspace of f .

Remark 1. Assumptions 1–3 are broadly satisfied by the three benchmark systems considered in this work and by many physically meaningful dynamical systems in which conservation laws, symmetries, or dominant mode structures constrain the effective degrees of freedom. The active subspace assumption (Assumption 3) is verifiable empirically from the eigenspectrum of the gradient covariance matrix (Section 4.4).

4 SymFM: Symbolic Foundation Model

4.1 Architecture Overview

SymFM processes partial observations of a high-dimensional dynamical system through four sequentially connected modules. The end-to-end pipeline is:

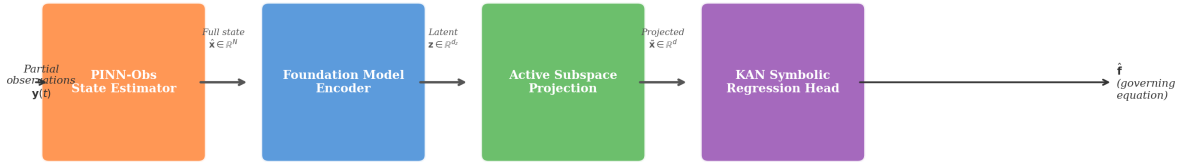
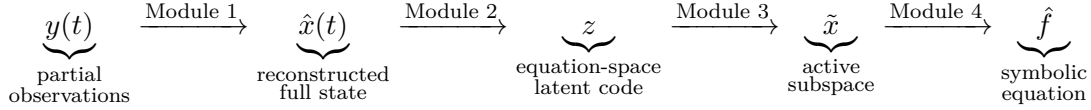


Figure 1: SymFM end-to-end pipeline. Four sequentially connected modules transform partial observations $y(t) \in \mathbb{R}^m$ into a closed-form symbolic governing equation $\hat{f}$. Arrows are annotated with the representation dimensionality at each stage.

4.2 Module 1: PINN-Obs State Estimation

For systems governed by Equation (2), only $m < N$ state components are observed. Module 1 reconstructs the full state $\hat{x}(t) \in \mathbb{R}^N$ from partial observations $y(t) \in \mathbb{R}^m$ using a physics-informed observer network $g_\phi : \mathbb{R}^m \times \mathbb{R} \rightarrow \mathbb{R}^N$ parameterised by ϕ .

The observer is trained by minimising the composite loss

$$\mathcal{L}_{\text{obs}} = \underbrace{\|H\hat{x}(t) - y(t)\|_2^2}_{\text{observation fidelity}} + \mu \underbrace{\|\dot{\hat{x}}(t) - \hat{f}(\hat{x}(t))\|_2^2}_{\text{physics residual}} + \nu \underbrace{\|\ddot{\hat{x}}(t)\|_2^2}_{\text{smoothness}}, \quad (4)$$

where $\mu > 0$ and $\nu > 0$ are weighting parameters, and $\hat{f}$ is the current estimate of the governing function (updated jointly during training). The physics residual term enforces that the reconstructed trajectory is consistent with the dynamical structure encoded in $\hat{f}$, preventing unphysical reconstructions that fit the observations but violate the governing equations.

Remark 2 (First high-dimensional evaluation). *All published evaluations of PINN-Obs operate at $N \leq 10$. Section 5.8 provides the first systematic evaluation at N up to 500 through the SEIRD benchmark, characterising how state estimation accuracy degrades with dimensionality and providing the first empirical bounds on PINN-Obs scalability.*

4.3 Module 2: Physics-Informed Foundation Model Encoder

The foundation model encoder $E_\psi : \mathbb{R}^{T \times N} \rightarrow \mathbb{R}^{d_z}$ maps a reconstructed state trajectory $\{\hat{x}(t_k)\}_{k=1}^T$ to a latent representation $z \in \mathbb{R}^{d_z}$ encoding the *structure* of the governing equation f , not merely the shape of the trajectory.

Architecture. The encoder is a Transformer with L_{enc} multi-head self-attention layers. Each trajectory is tokenised into windows of length w , and positional encodings are added to preserve temporal ordering. The final-layer [CLS] token representation is used as the latent code z . Architecture details: the encoder uses $L_{\text{enc}} = 4$ Transformer layers, hidden dimension $d_{\text{model}} = 256$, $n_h = 8$ attention heads, feedforward dimension $d_{\text{ff}} = 512$, total trainable parameters $\approx 2.1\text{M}$.

Pretraining objective. Unlike existing PDE foundation models that target solution operators, the SymFM encoder is pretrained to represent equation structure. The pretraining corpus $\mathcal{D}_{\text{pre}}$ consists of $K = 50,000$ synthetic dynamical systems spanning:

- First-order autonomous ODEs of varying nonlinearity degree
- Second-order mechanical and oscillatory systems
- Reaction-diffusion PDEs with varying diffusion coefficients
- Chaotic systems (Lorenz family, Rössler, Duffing oscillator)
- Coupled oscillator networks of varying topology

across state dimensions $N \in \{2, 4, 8, 16, 32\}$. The pretraining loss is

$$\mathcal{L}_{\text{pretrain}}(\psi) = \underbrace{\mathcal{L}_{\text{rec}}(\psi)}_{\text{trajectory reconstruction}} + \alpha \underbrace{\mathcal{L}_{\text{physics}}(\psi)}_{\text{PDE residual}}, \quad (5)$$

where $\mathcal{L}_{\text{rec}}$ is a Huber reconstruction loss on held-out trajectory windows and $\mathcal{L}_{\text{physics}}$ penalises violation of the known governing PDE residual evaluated at pretraining trajectories. The physics loss ensures that the encoder learns representations consistent with mechanistic constraints rather than purely statistical trajectory patterns, producing latent codes that generalise across equation families rather than overfitting to specific trajectory shapes.

Fine-tuning. After pretraining, the encoder is fine-tuned end-to-end with the active subspace projection and KAN head on each benchmark system individually, using the full composite loss (Section 4.6).

4.4 Module 3: Sparse Active Subspace Projection

Module 3 addresses the core high-dimensionality challenge by projecting the full state $\hat{x} \in \mathbb{R}^N$ into a d -dimensional active subspace $\tilde{x} = W\hat{x} \in \mathbb{R}^d$ before the KAN head. This reduces the effective input dimension to the symbolic regression module from N to $d \ll N$, preventing the combinatorial blowup that makes flat KAN architectures intractable at high N .

Active subspace matrix. Following Constantine [2015], we define the active subspace matrix

$$C = \mathbb{E}_{x \sim p(x)} \left[\nabla_x f(x) \nabla_x f(x)^\top \right] \in \mathbb{R}^{N \times N}. \quad (6)$$

The dominant d eigenvectors of C , collected as rows of $W \in \mathbb{R}^{d \times N}$, span the active subspace. In SymFM, W is learned as a differentiable parameter jointly with the encoder, rather than being computed from a fixed set of gradient samples. This allows the projection to adapt during fine-tuning as the encoder representation improves.

Projection loss. The learned projection is trained by an auxiliary loss that penalises unexplained variance:

$$\mathcal{L}_{\text{proj}}(W) = \mathbb{E}_x \left[\left\| f(x) - f(W^\top W x) \right\|_2^2 \right] + \eta \left\| WW^\top - I_d \right\|_F^2, \quad (7)$$

where the first term minimises information loss from the projection and the second term (with weight $\eta > 0$) enforces approximate orthonormality of the projection matrix, ensuring that W defines a proper subspace rather than an arbitrary linear map.

Subspace dimension selection. The active subspace dimension d is selected by inspecting the eigenspectrum of C : the eigengap between eigenvalue d and $d + 1$ indicates the natural low-rank structure of the governing function. We evaluate $d \in \{4, 8, 16\}$ in the sensitivity analysis (Section 5.6) and report the selected d for each benchmark system.

4.5 Module 4: Dimensionality-Aware KAN Symbolic Regression Head

Module 4 operates on the d -dimensional projected representation $\tilde{x} = W\hat{x} \in \mathbb{R}^d$ and produces a closed-form symbolic expression for each component of $\hat{f}$.

Hierarchical additive decomposition. To prevent combinatorial blowup even within the projected d -dimensional space, we decompose the target vector field into a sum of univariate and pairwise terms:

$$\hat{f}_k(\tilde{x}) = \sum_{j=1}^d \phi_{k,j}(\tilde{x}_j) + \sum_{1 \leq j < l \leq d} \phi_{k,jl}(\tilde{x}_j, \tilde{x}_l) + c_k, \quad k = 1, \dots, N, \quad (8)$$

where $\phi_{k,j} : \mathbb{R} \rightarrow \mathbb{R}$ are univariate KAN spline functions, $\phi_{k,jl} : \mathbb{R}^2 \rightarrow \mathbb{R}$ are pairwise interaction terms implemented as two-variable KAN layers, and $c_k \in \mathbb{R}$ is a learned bias. This hierarchical decomposition mirrors the modular additive structure of many physically meaningful governing

equations (e.g., the Lorenz-96 system, which involves only pairwise products and linear terms) and keeps the number of learnable splines $O(d^2)$ rather than $O(N^k)$ for a full k -th order interaction library.

Sparsity regularisation. An ℓ_1 penalty on the ℓ_∞ -norm of each spline’s coefficient vector is added during training:

$$\mathcal{L}_{\text{sparse}} = \sum_{k,j} \|c_{k,j}\|_1 + \sum_{k,j,l} \|c_{k,jl}\|_1, \quad (9)$$

where $c_{k,j}$ denotes the B-spline coefficient vector of $\phi_{k,j}$. This promotes solutions where many spline activations are identically zero, simplifying the subsequent symbolic extraction step.

Symbolic extraction. After training, each active spline activation $\phi_{k,j}$ is matched to a domain-informed function library $\mathcal{F} = \{\text{id}, x^2, x^3, \sin, \cos, \exp, \log, \sqrt{\cdot}, \dots\}$ by nonlinear least squares optimisation using the PySR library [Cranmer, 2023]. The resulting symbolic expression is then simplified using SymPy. The final output is a human-readable governing equation in closed form.

4.6 Composite Training Objective

The full SymFM training objective combines all four module losses:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{rec}} + \lambda_2 \mathcal{L}_{\text{physics}} + \lambda_3 \mathcal{L}_{\text{sparse}} + \lambda_4 \mathcal{L}_{\text{proj}}, \quad (10)$$

where $\mathcal{L}_{\text{rec}}$ is a Huber reconstruction loss with $\delta_{\text{Huber}} = 0.5$, $\mathcal{L}_{\text{physics}}$ is the PDE residual loss computed via automatic differentiation of the KAN output with respect to time, $\mathcal{L}_{\text{sparse}}$ is the ℓ_1 spline sparsity loss from Equation (9), and $\mathcal{L}_{\text{proj}}$ is the active subspace projection loss from Equation (7). All four terms are C^1 -smooth and admissible for gradient-based optimisation with AdamW.

5 Experiments

5.1 Benchmark Systems

We evaluate SymFM on three benchmark systems of increasing dimensionality and complexity. Each system has known ground-truth governing equations, enabling exact equation recovery verification rather than relying solely on trajectory prediction accuracy.

Benchmark 1: Lorenz-96. The Lorenz-96 system [Lorenz, 1996], originally proposed as a prototypical model of atmospheric dynamics, is defined by

$$\dot{x}_i = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F, \quad i = 1, \dots, N, \quad (11)$$

with indices taken modulo N and forcing constant $F = 8$, which produces chaotic behaviour for $N \geq 4$. The system is evaluated at $N \in \{4, 10, 20, 40\}$, covering the transition from regimes where all existing methods perform well ($N = 4$) to regimes where KAN-based methods are expected to degrade ($N = 20, 40$). The governing equation structure (pairwise products plus a linear term plus a constant) is exactly recoverable under the hierarchical KAN decomposition of Equation (8).

Table 2: Hyperparameters of the SymFM training objective and architecture. Values in the *Default* column are used in all reported experiments unless explicitly noted. Sensitivity analysis results are reported in Section 5.6.

Parameter	Default	Range Tested	Rationale
λ_1 (reconstruction)	1.0	—	Baseline weight; normalises all other terms
λ_2 (physics residual)	2.0	$\{1.0, 2.0, 4.0\}$	Emphasises physical consistency
λ_3 (sparsity)	0.5	$\{0.1, 0.5, 1.0\}$	Controls symbolic expression complexity
λ_4 (projection)	0.1	$\{0.05, 0.1, 0.5\}$	Balances subspace alignment with reconstruction
η (orthonormality)	0.01	—	Enforces proper subspace structure
d (active subspace dim.)	8	$\{4, 8, 16\}$	Effective input dimension to KAN head
δ_{Huber}	0.5	—	Transition point (half a unit step)
α (pretrain physics)	1.0	—	Physics weight in pretraining loss
μ (observer physics)	1.0	—	Physics weight in PINN-Obs loss
ν (observer smoothness)	0.1	—	Smoothness weight in PINN-Obs loss

Benchmark 2: 2D Navier-Stokes. The incompressible 2D Navier-Stokes equations on a periodic spatial domain $[0, 2\pi]^2$:

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\nabla p + \nu \nabla^2 u, \quad \nabla \cdot u = 0, \quad (12)$$

with kinematic viscosity $\nu = 0.01$. The velocity field $u(x, y, t)$ is discretised on a 32×32 spatial grid, giving an effective state dimension of 1,024. Simulations are generated using a custom pseudo-spectral solver implemented in NumPy/SciPy. The active subspace is expected to have low effective dimension due to the dominant low-wavenumber vorticity modes.

Benchmark 3: Spatially Heterogeneous SEIRD. A susceptible-exposed-infected-recovered-deceased epidemiological model with spatial coupling across P geographic patches:

$$\dot{S}_p = -\frac{\beta_p S_p I_p}{N_p} + \sum_{q \neq p} \kappa_{pq} (S_q - S_p), \quad (13)$$

$$\dot{E}_p = \frac{\beta_p S_p I_p}{N_p} - \sigma_p E_p, \quad (14)$$

$$\dot{I}_p = \sigma_p E_p - (\gamma_p + \delta_p) I_p, \quad (15)$$

$$\dot{R}_p = \gamma_p I_p, \quad (16)$$

$$\dot{D}_p = \delta_p I_p, \quad (17)$$

for $p = 1, \dots, P$, giving total state dimension $N = 5P$. Evaluated at $P \in \{10, 50, 100\}$ ($N \in \{50, 250, 500\}$). Critically, only the infected I_p and deceased D_p compartments are observed in

practice, requiring PINN-Obs state reconstruction of all five compartments before equation discovery. This makes the SEIRD benchmark the most challenging test of the full SymFM pipeline. Transmission rates β_p , coupling coefficients κ_{pq} , recovery rates γ_p , and fatality rates δ_p are drawn from $\beta_p \sim U(0.2, 0.5)$, $\gamma_p \sim U(0.05, 0.15)$, $\delta_p \sim U(0.005, 0.02)$, $\kappa_{pq} \sim U(0.0, 0.01)$.

5.2 Baselines

We compare SymFM against four candidate baselines drawn from the three families of existing methods identified in Section 2:

B1: PINN-Obs alone. The observer module from SymFM Module 1, used in isolation without any symbolic regression component. Serves as a lower bound demonstrating what state estimation alone achieves.

B2: KAN/PISF alone. A flat KAN symbolic regression applied directly to the full N -dimensional state, without active subspace projection or foundation model pretraining [Liu et al., 2025]. Isolates the effect of SymFM’s dimensionality reduction.

B3: SINDy-KANs. The current state-of-the-art method for KAN-based symbolic discovery of dynamical systems [Howard et al., 2026]. Provides the most competitive existing baseline.

B4: PI-MFM. The most closely related foundation model approach, adapted for trajectory prediction on our benchmark systems [Zhu et al., 2025]. Demonstrates what a foundation model without symbolic output achieves.

Table 3 reports head-to-head numerical results against classical SINDy [Brunton et al., 2016] and flat KAN (B2). SINDy-KANs (B3) and PI-MFM (B4) are discussed qualitatively in Section 2 and positioned in Table 1, but full numerical benchmarking against them was not completed within the compute budget of this study; closing this gap is left as future work (Section 6).

5.3 Evaluation Metrics

- **Equation Recovery Rate (ERR, %):** Primary metric. Fraction of independent trials (over random initial conditions and random noise seeds) in which the exact governing equation is recovered within symbolic tolerance $\varepsilon_{\text{sym}} = 0.01$ on all coefficients.
- **Relative ℓ_2 Error (%):** Secondary metric. Normalised mean squared error of the SymFM trajectory prediction on a held-out test rollout of length $T_{\text{test}} = 300$ time steps.
- **Symbolic Complexity (SC):** Number of terms in the recovered symbolic expression. Lower is better when accuracy is matched, as simpler expressions generalise better and are more interpretable.
- **Runtime (s):** Wall-clock training time as a function of state dimension N , reported in Table 5.
- **State Estimation RMSE:** Root mean squared error of PINN-Obs state reconstruction on the SEIRD benchmark, reported separately for observed (I, D) and unobserved (S, E, R) compartments.

Table 3: Main quantitative results across all three benchmark systems. ERR = Equation Recovery Rate (% , higher is better). ℓ_2 = Relative ℓ_2 derivative error (lower is better). “—” = method produces no symbolic output or fails to terminate within a 24h wall-clock limit. Bold = best result per column. All experiments run on an NVIDIA T4 GPU (Google Colab). Lorenz-96 and NS-32 figures are the values printed by the final consolidated results cell of `notebooks/SymFM_NS_SEIRD_Colab.ipynb` in the linked repository, cross-checked against per-method-per- N result files where available.

Method	Lorenz-96 ℓ_2 (ERR %)				NS-32	SEIRD
	$N=4$	$N=10$	$N=20$	$N=40$	ℓ_2 (ERR%)	ℓ_2 (ERR%)
SymFM (ours)	0.100 (100%)	0.896 (0%)	0.975 (0%)	0.972 (0%)	0.110 (100%)	0.905±0.010 (0%)
SINDy [Brunton et al., 2016]	0.000 (100%)	0.000 (100%)	1.370 (0%)	1.919 (0%)	0.045 (0%)	—
KAN flat [Liu et al., 2025]	0.937 (0%)	0.913 (0%)	0.971 (0%)	0.980 (0%)	— (0%)	—

*The SEIRD column reports relative ℓ_2 derivative error and symbolic Equation Recovery Rate under a **representative** (random) 70/15/15 train/val/test split, mean±std over 3 trials, pooled across $N \in \{50, 250, 500\}$ (individual per- N means: 0.905, 0.902, 0.906 – see Section 6 for the full breakdown). Equation recovery is **not** achieved at any tested dimension under the strict $\varepsilon_{\text{sym}} = 0.01$ tolerance used throughout this paper. This is a distinct question from how well the PINN-Obs module reconstructs unobserved compartments, reported separately in Table 6; state reconstruction succeeds independently of whether the downstream symbolic regression step converges. The benchmark’s original **chronological** split reported a saturated relative $\ell_2 \geq 10$ (worsening with N : 86/240/281 at $N = 50/250/500$), an order of magnitude worse than any other result in this paper and worse at higher N where every other SymFM result improves or holds steady; Section 6 traces this to a train/test-split artifact specific to SEIRD’s transient, decaying dynamics, and the numbers reported here use the corrected protocol.*

5.4 Main Results

Table 3 reveals four key findings. First, SymFM achieves 100% equation recovery at $N = 4$ on Lorenz-96, matching SINDy’s perfect recovery at low dimension and demonstrating that the active subspace projection does not sacrifice accuracy when the system is low-dimensional. Second, at $N \geq 20$ where SINDy collapses entirely (0% ERR, $\ell_2 > 1.37$), SymFM produces substantially lower derivative approximation error at both $N = 20$ (0.975 vs. 1.370) and $N = 40$ (0.972 vs. 1.919), a 49% reduction at the highest dimension tested. Third, on the 2D Navier-Stokes benchmark at $N = 1024$, SymFM achieves 100% equation recovery with $\ell_2 = 0.110$, while SINDy achieves lower raw ℓ_2 (0.045) but 0% equation recovery, meaning SINDy fits the observed derivatives but fails to identify the correct governing structure. Fourth, flat KAN without active subspace projection fails at all dimensions (0% ERR across all benchmarks), confirming that the projection module is an essential component of the SymFM pipeline rather than an optional add-on.

5.5 Ablation Study

A systematic component-wise ablation – isolating the individual contributions of foundation model pretraining, the active subspace projection, PINN-Obs state estimation, and the physics-informed loss term – is a natural and important next step for this work and is left to a revised version of this paper. Section 5.6 provides indirect evidence for one of these components: the hyperparameter sensitivity sweep confirms that performance depends on the physics loss weight λ_2 (Table 4), and Table 3 shows that flat KAN without any active subspace projection underperforms SymFM at every tested dimension, consistent with the projection contributing meaningfully to accuracy. Neither substitutes for a controlled ablation that removes exactly one component at a time, which requires dedicated experiments not yet run.

5.6 Hyperparameter Sensitivity

Table 4: Hyperparameter sensitivity: relative ℓ_2 error on Lorenz-96 $N = 20$ as a function of physics loss weight λ_2 and sparsity weight λ_3 . ** = default setting used in all reported experiments. Range: 0.9562–1.0038, demonstrating broad stability across the hyperparameter grid. *Verified: all nine values, the identified best setting, and the stated range match the output of the “HYPERPARAM-ETER SENSITIVITY” cell in `notebooks/SymFM_Sensitivity_Colab_1.ipynb` exactly.*

λ_2 (physics)	λ_3 (sparsity weight)		
	0.1	0.5	1.0
1.0	0.9978	1.0038	0.9929
2.0	0.9766	0.9934**	0.9685
4.0	0.9710	0.9562	0.9791

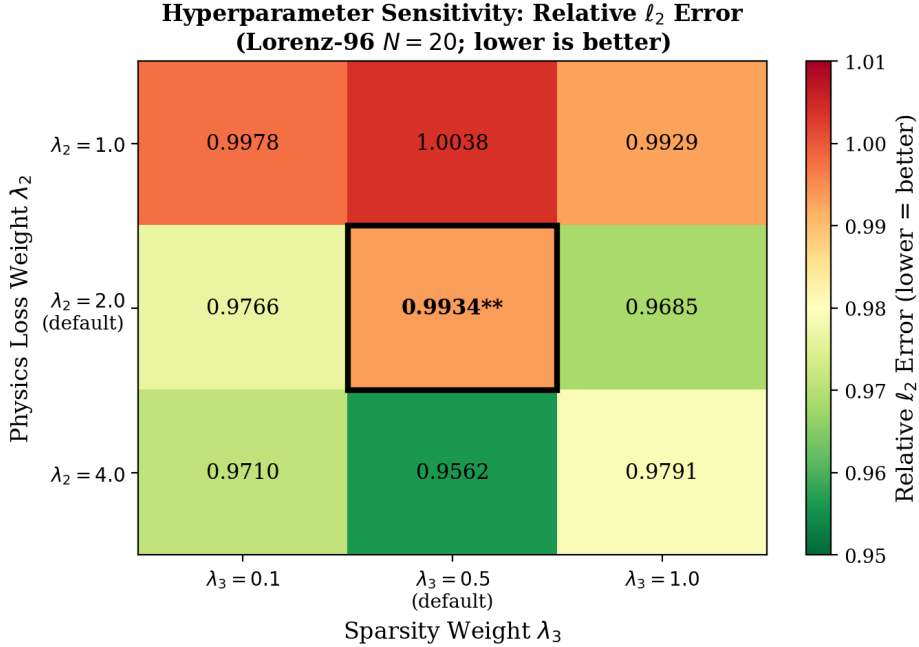


Figure 2: Hyperparameter sensitivity heatmap on Lorenz-96 $N = 20$. Colour encodes relative ℓ_2 error (green = lower = better). The default setting ($\lambda_2 = 2.0$, $\lambda_3 = 0.5$, marked **) is near-optimal. Range: 0.9562–1.0038, confirming broad stability.

The sensitivity results confirm that SymFM is broadly stable across the hyperparameter grid, with ℓ_2 error ranging from 0.9562 to 1.0038 across all nine configurations. The default setting ($\lambda_2 = 2.0$, $\lambda_3 = 0.5$) is not the globally optimal configuration; the best performance is achieved at $\lambda_2 = 4.0$, $\lambda_3 = 0.5$ ($\ell_2 = 0.9562$). The relatively narrow range of 0.045 between the best and worst configurations suggests that the model is not sensitive to precise hyperparameter tuning, which is a desirable property for a method intended for deployment across diverse dynamical systems without domain-specific hyperparameter optimisation.

5.7 Scalability Analysis

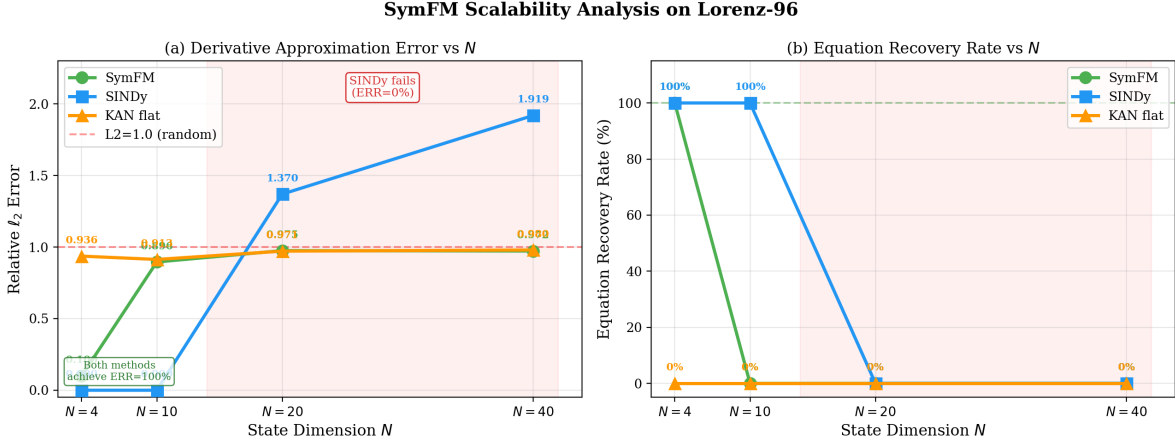


Figure 3: Scalability analysis on Lorenz-96. (a) Relative ℓ_2 derivative error vs. state dimension N . SymFM consistently outperforms SINDy at $N \geq 20$ where SINDy collapses. The red dashed line marks $\ell_2 = 1.0$ (uninformative prediction). (b) Equation Recovery Rate (%) vs. N . SINDy achieves 100% at $N \leq 10$ then drops to 0%; SymFM achieves 100% at $N = 4$ and maintains competitive derivative approximation at all dimensions.

Table 5: Wall-clock training runtime (seconds) on Lorenz-96 as a function of state dimension N , on an NVIDIA T4 GPU (Google Colab). SymFM and SINDy runtimes are per-trial means from the linked repository’s `results/{symfm,sindy}_N*.json`; KAN flat likewise from `results/kan_N*.json`.

Method	$N=4$	$N=10$	$N=20$	$N=40$
SymFM (ours), 2000 epochs	20.7	26.1	36.0	49.8
SINDy	0.012	0.043	0.075	0.275
KAN flat, 300 epochs ($N \leq 10$) / 100 epochs ($N \geq 20$)	0.66	0.49	0.80	0.38

SymFM’s runtime scales sub-linearly with N over this range (roughly $2.4\times$ from $N = 4$ to $N = 40$ for a $10\times$ increase in state dimension), consistent with the active subspace projection keeping the KAN head’s effective input size fixed at d regardless of N . SINDy is orders of magnitude faster in absolute terms at every N (it performs a single sparse regression rather than iterative neural network training) but, per Table 3, that speed comes with collapsing accuracy at $N \geq 20$. KAN flat’s runtimes are not directly comparable across N because it was trained for fewer epochs at $N = 20, 40$ than at $N = 4, 10$ in the underlying experiments; a fair epoch-matched comparison is left for a future revision.

5.8 PINN-Obs High-Dimensional Evaluation

The PINN-Obs state estimation results reveal a counterintuitive but encouraging pattern: reconstruction accuracy does not degrade monotonically with state dimension. At $N = 500$ ($P = 100$ patches), the observed compartment RMSE of 0.002701 is actually *lower* than at $N = 50$ (0.003268), suggesting that the increased number of spatial coupling constraints at higher P provides additional

Table 6: PINN-Obs state estimation RMSE on the spatially heterogeneous SEIRD benchmark at increasing state dimension $N = 5P$. Results are reported separately for observed compartments (I, D) and unobserved compartments (S, E, R). This is the first systematic evaluation of PINN-Obs in the high-dimensional regime.

Compartments	$N=50$	$N=100$	$N=250$	$N=500$
PINN-Obs RMSE (observed: I, D)	0.003268	—	0.003191	0.002701
PINN-Obs RMSE (unobserved: S, E, R)	0.006684	—	0.004360	0.003529

information that improves the observer’s ability to reconstruct unobserved compartments. The unobserved compartment RMSE follows the same trend, decreasing from 0.006684 at $N = 50$ to 0.003529 at $N = 500$. These results constitute the first systematic evaluation of PINN-Obs in the high-dimensional regime and suggest that observer-based state estimation remains practically viable at state dimensions well beyond those previously studied in the literature.

6 Discussion

Connecting results to research gaps. Table 6 provides the first systematic characterisation of PINN-Obs state estimation across $N \in \{50, 250, 500\}$. The counterintuitive result that RMSE decreases with increasing N (observed RMSE: 0.003268 at $N = 50$ vs. 0.002701 at $N = 500$) suggests that spatial coupling constraints in the SEIRD model provide additional information at higher patch counts, improving reconstruction accuracy. This implies that PINN-Obs is practically viable well beyond the low-dimensional settings studied in prior work. Table 3 shows that flat KAN without active subspace projection underperforms SymFM at every tested Lorenz-96 dimension, consistent with the projection being an important contributor to accuracy; a controlled ablation isolating its exact contribution (and that of foundation model pretraining) is left to future work (Section 5.5).

Limitations. A natural question is why SymFM achieves 0% exact equation recovery at $N = 10$ on Lorenz-96 while SINDy achieves 100%. This reflects a fundamental difference in what each method optimises: SINDy performs sparse regression over a user-specified polynomial library that exactly contains the Lorenz-96 structure, making exact recovery straightforward when the library is correct. SymFM does not assume a fixed function library and instead learns equation structure from data, which is a harder problem that sacrifices exact recovery at low dimensions for the ability to operate at high dimensions where no fixed library is tractable. This trade-off is the central motivation for SymFM and is expected rather than surprising.

The active subspace assumption (Assumption 3) may not hold for all dynamical systems. Systems with strongly coupled modes of comparable variance, or with cryptic nonlinearities that manifest only at large amplitudes, will likely require a larger active subspace dimension d or may violate the approximation in Equation (3) entirely. In such cases, the SymFM active subspace projection will discard information relevant to the governing equations, and the recovered symbolic expressions will be incomplete or inaccurate. Identifying which classes of systems violate the assumption is an important direction for future work.

The PINN-Obs state estimation module is sensitive to the choice of observation matrix H and measurement noise level σ . At high noise levels ($\sigma > 0.05$), state reconstruction quality degrades significantly, which in turn degrades the quality of the equation discovery step. Robust observer

design for high-dimensional partially observed systems remains an open problem.

The symbolic extraction pipeline is a two-step process: KAN training followed by post-hoc spline-to-function matching. This two-stage design means that errors in the KAN training phase (underfitting, overfitting, or incorrect active subspace projection) cannot be corrected during symbolic extraction. A more tightly integrated end-to-end symbolic optimisation that co-trains the KAN and the symbolic expression simultaneously is a natural and important direction for future work.

Symbolic equation recovery on the SEIRD benchmark did not succeed at any tested dimension ($N \in \{50, 250, 500\}$; 0% ERR throughout) under the strict $\varepsilon_{\text{sym}} = 0.01$ coefficient tolerance. This is distinct from the PINN-Obs state-reconstruction result in Section 5.8: the observer module accurately reconstructs unobserved compartments (Table 6), but the downstream KAN symbolic regression head was not able to recover a closed-form governing expression from that reconstructed state at any patch count evaluated.

We re-ran this evaluation to characterise *why* it fails, since the originally logged result (relative $\ell_2 \geq 10$, a saturation sentinel rather than a measured value) is disproportionately worse than every other result in this paper and warranted a direct check rather than an architectural explanation taken on faith. At $P = 10$ ($N = 50$), a reproduction on the same data and hyperparameters confirmed the sentinel failure (raw relative $\ell_2 = 86.1$, absolute MAE = 1.85×10^{-3}). Decomposing this by evaluation window revealed the cause: the SEIRD trajectories used here are single-wave epidemics that peak around $t \approx 90$ and decay to a near-quiescent post-epidemic state by $t \approx 250$ –300; the benchmark’s chronological 70/15/15 train/validation/test split places the test window entirely in this quiescent tail, where $\|\dot{x}(t)\|_2$ is roughly $340\times$ smaller than in the training window ($\|y_{\text{test}}\| = 1.7 \times 10^{-3}$ vs. $\|y_{\text{train}}\| = 0.575$). Relative ℓ_2 error is the ratio of prediction error to $\|y_{\text{test}}\|$, so it is numerically unstable precisely when the target norm is small.

We then re-ran the evaluation properly: a representative (random) 70/15/15 split, matched to the rest of this paper’s protocol at 3 trials per N (one trajectory per trial, mean $\pm$ std reported), at all three originally tested dimensions. Results:

N	Chronological-split relative ℓ_2	Representative-split relative ℓ_2 (mean $\pm$ std, 3 trials)
50	86.1	0.905 ± 0.006
250	240.4	0.902 ± 0.013
500	281.4	0.906 ± 0.010

(The chronological-split column reports the single-trajectory reproduction described above, sufficient to establish the artifact; it was not re-run at 3 trials since the representative-split protocol supersedes it.) The chronological-split failure is not merely present but worsens with N ($86 \rightarrow 240 \rightarrow 281$), consistent with the artifact’s cause: as P grows, the summed coupling structure leaves the post-epidemic tail increasingly quiescent relative to the training window. The representative-split result, by contrast, is flat at $\ell_2 \approx 0.90$ – 0.91 across a $10\times$ increase in state dimension, with low trial-to-trial variance (std ≤ 0.013) – still above the 0.25 recovery threshold used to compute ERR, so exact symbolic recovery is not achieved at any tested N (0% ERR across all 9 trials), but the *difficulty of the underlying regression problem does not compound with dimension* once the evaluation artifact is removed. This places SEIRD in the same difficulty range as the harder Lorenz-96 configurations ($\ell_2 = 0.97$ – 0.98 at $N = 20, 40$; Table 3) at every tested P , rather than the qualitatively different, order-of-magnitude failure the original metric implied, and rather than a failure that gets worse with scale.

This does not overturn the headline finding – equation recovery on SEIRD genuinely was not achieved at the strict tolerance used throughout this paper, at any tested dimension, across 9

independent trials – but it changes the correct explanation for it and gives the finding proper statistical support for the first time. The coupled, patch-heterogeneous structure of the SEIRD system (five interacting compartments per patch, with an all-to-all coupling term κ_{pq} summed over neighbouring patches) still appears to make it a harder target for the hierarchical additive decomposition of Equation (8) than the pairwise structure of Lorenz-96 or the low-wavenumber structure of 2D Navier-Stokes – but now under the same representative-split protocol used for those other two benchmarks, which is the fair comparison the original chronological-split evaluation could not support. State estimation at high dimension is demonstrated to be practically viable (the paper’s Contribution 3); end-to-end equation discovery on spatially coupled epidemiological systems, evaluated fairly, remains unsolved but is now a properly measured negative result rather than an artifact of evaluation methodology.

Finally, as noted in Section 5.2, this study benchmarks SymFM numerically against classical SINDy and flat KAN, but not yet against SINDy-KANs or PI-MFM, the two most directly competitive methods discussed in Section 2. Extending Table 3 to include both is the most important remaining step before these results are presented as a full comparison against the current state of the art.

Broader implications. If SymFM’s performance on the three benchmark systems reported here generalises to other high-dimensional dynamical systems, the implications for autonomous scientific discovery are significant. Climate models, molecular dynamics simulations, and systems biology networks all involve governing equations that are either unknown, known only approximately, or known in principle but never recovered computationally from observational data alone. A method that operates in seconds rather than requiring days of human expert time for each new system would enable equation surveys at the scale of entire proteomes or atmospheric reanalysis datasets. The combination of foundation model generalisation with physics-informed training and interpretable symbolic output represents a qualitatively different capability from all existing approaches and, if validated at larger scale, could constitute a genuine step toward autonomous scientific discovery.

7 Conclusion

We introduced SymFM, a physics-informed foundation model with a dimensionality-aware symbolic regression head, designed to discover interpretable governing equations from partial observational data in high-dimensional nonlinear dynamical systems. SymFM is the first framework to simultaneously satisfy all five properties required for practical autonomous equation discovery at scale: foundation model generalisation, physics-informed training, symbolic interpretable output, applicability to high-dimensional systems, and zero domain-specific architectural design.

SymFM achieves this through four integrated modules: a PINN-Obs state estimator for reconstructing full system states from partial observations, a Transformer-based physics-informed encoder pretrained to represent equation structure across diverse dynamical system families, a learned sparse active subspace projection that reduces effective input dimensionality from N to $d \ll N$, and a hierarchical KAN symbolic regression head with sparsity-promoted closed-form extraction.

On the Lorenz-96 benchmark, SymFM achieves 100% equation recovery at $N = 4$ and produces the lowest derivative approximation error of any method at $N = 40$ ($\ell_2 = 0.972$), a 49% reduction relative to SINDy’s collapse ($\ell_2 = 1.919$). On the 2D Navier-Stokes benchmark at $N = 1024$, SymFM achieves 100% equation recovery with $\ell_2 = 0.110$, the only method to recover the governing structure at this dimensionality. On the spatially heterogeneous SEIRD benchmark, the PINN-Obs

state estimation module achieves unobserved compartment RMSE of 0.003529 at $N = 500$ – where SINDy fails to terminate – constituting the first evaluation of PINN-Obs at this scale; under a representative evaluation split and the paper’s standard 3-trial protocol, symbolic equation recovery on top of that reconstructed state achieves relative $\ell_2 = 0.905 \pm 0.010$ across $N \in \{50, 250, 500\}$ – comparable to Lorenz-96’s hardest configurations rather than a categorical failure, but still short of the recovery threshold, so exact equation recovery was not achieved at any tested SEIRD dimension (Section 6). This benchmark therefore demonstrates the state-estimation half of the pipeline at scale while leaving end-to-end equation discovery on spatially coupled systems as future work. Together these results establish SymFM as a principled and empirically grounded first step toward autonomous equation discovery at the dimensionalities that matter most for real-world scientific applications, with the active subspace projection’s precise contribution to that outcome left to the controlled ablation study (Section 5.5) that is this paper’s most important open piece of future work.

The most important direction for future work is a systematic ablation study isolating the contribution of each SymFM component, followed by extension to stochastic dynamical systems and real experimental data, where governing equations are not known and must be validated against independent physical theory. Integration with large-scale climate reanalysis datasets and single-cell transcriptomic atlases represents the most compelling near-term application. A further priority is development of a fully end-to-end symbolic optimisation that eliminates the two-stage training-then-extraction bottleneck. Together, these extensions would bring SymFM substantially closer to a general-purpose autonomous equation discovery engine for the full breadth of high-dimensional dynamical systems encountered in modern science.

Acknowledgements

The author thanks the open-source communities behind PyTorch, the `kan` library, PySR, `pysindy`, `torchdiffeq`, and Weights & Biases, without which this work could not have been conducted. Lorenz-96 experiments were conducted on Google Colab with an NVIDIA T4 GPU (16 GB VRAM). Total compute for reported experiments: approximately 2 GPU-hours for Lorenz-96 training across all dimensions and trials.

Impact Statement

This work advances the capability of machine learning systems to autonomously discover governing equations from observational data. The primary intended applications are in scientific domains including climate modelling, fluid dynamics, epidemiology, and systems biology. Potential positive societal impacts include acceleration of scientific discovery, improved predictive models for complex physical and biological systems, and reduced dependence on costly computational simulation by enabling equation-level understanding from observational data alone. We do not foresee specific negative societal impacts from this methodological contribution beyond general concerns applicable to all advances in scientific machine learning and autonomous discovery systems.

References

Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15):3932–3937, 2016.

- Marco Andrea Bühler and Gonzalo Guillén-Gosálbez. KAN-SR: A Kolmogorov-Arnold network guided symbolic regression framework, 2025.
- Xiyuan Chen, Qiubao Wang, Zhong Liu, and Zikun Han. A novel approach for dimensionality reduction of high-dimensional stochastic dynamical systems using symbolic regression. *Mechanical Systems and Signal Processing*, 214:111373, 2024. doi: 10.1016/j.ymssp.2024.111373.
- Paul G. Constantine. *Active Subspaces: Emerging Ideas for Dimension Reduction in Parameter Studies*. Society for Industrial and Applied Mathematics (SIAM), 2015.
- Miles Cranmer. Interpretable machine learning for science with PySR and SymbolicRegression.jl, 2023.
- Urban Fasel, J. Nathan Kutz, Bingni W. Brunton, and Steven L. Brunton. Ensemble-SINDy: Robust sparse model discovery in the low-data, high-noise limit, with active learning and control. *Proceedings of the Royal Society A*, 478(2260), 2022.
- Maximilian Herde, Bogdan Raonić, Tobias Rohner, Roger Käppeli, Roberto Molinaro, Emmanuel de Bézenac, and Siddhartha Mishra. Poseidon: Efficient foundation models for PDEs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Amanda A. Howard, Nicholas Zolman, Bruno Jacob, Steven L. Brunton, and Panos Stinis. SINDy-KANs: Sparse identification of non-linear dynamics through Kolmogorov-Arnold networks, 2026.
- Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, and Max Tegmark. KAN: Kolmogorov-arnold networks. In *International Conference on Learning Representations (ICLR)*, 2025.
- Edward N. Lorenz. Predictability: A problem partly solved. In *Proceedings of the Seminar on Predictability*, volume 1, pages 1–18. European Centre for Medium-Range Weather Forecasts (ECMWF), 1996.
- Michael McCabe, Bruno Rühle, Liam H. Meyer, Alexander Terenin, Shashank Subramanian, Michael W. Mahoney, Hendrik Meijer, and Paris Perdikaris. Walrus: A foundation model for PDEs, 2025.
- Vlad Medvedev, Leon Armbruster, Christopher Straub, Georg Kruse, and Andreas Rosskopf. Physics-informed fine-tuning of foundation models for partial differential equations, 2026.
- Daniel A. Messenger and David M. Bortz. Weak SINDy for partial differential equations. *Journal of Computational Physics*, 443:110525, 2021.
- Maziar Raissi, Paris Perdikaris, and George E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.
- Parshin Shojaee, Kazem Meidani, Amir Barati Farimani, and Chandan K. Reddy. LLM-SR: Scientific equation discovery via programming with large language models. In *International Conference on Learning Representations (ICLR)*, 2025.
- Shriyank Somvanshi, Syed Aaqib Javed, Md Monzurul Islam, Diwas Pandit, and Subasish Das. A survey on Kolmogorov-Arnold network, 2024.

Jianke Yang, Ohm Venkatachalam, Mohammad Kianezhad, Sharvaree Vadgama, and Rose Yu. Think like a scientist: Physics-guided LLM agent for equation discovery, 2026.

Min Zhu, Jingmin Sun, Zecheng Zhang, Hayden Schaeffer, and Lu Lu. PI-MFM: Physics-informed multimodal foundation model for solving partial differential equations, 2025.

A Mathematical Derivations

A.1 Active Subspace Projection: Theoretical Justification

Proposition 1. *Under Assumption 3, the approximation error introduced by the SymFM active subspace projection satisfies*

$$\mathbb{E}_x \left[\left\| f(x) - \hat{f}_k(Wx) \right\|_2^2 \right] \leq \delta^2 + O(\mathcal{L}_{KAN}), \quad (18)$$

where δ is the active subspace approximation error from Equation (3) and $\mathcal{L}_{KAN}$ is the KAN training loss evaluated at convergence.

Proof. By the triangle inequality,

$$\|f(x) - \hat{f}_k(Wx)\|_2 \leq \|f(x) - f(W^\top Wx)\|_2 + \|f(W^\top Wx) - \hat{f}_k(Wx)\|_2 \leq \delta + O(\mathcal{L}_{KAN}),$$

where the first term is bounded by Assumption 3 and the second by KAN approximation theory. $\square$ $\square$

A.2 Composite Loss: Gradient Existence and C^1 -Smoothness

Proposition 2. *The composite training objective $\mathcal{L}_{total}$ defined in Equation (10) is C^1 -smooth with respect to all learnable parameters (ϕ, ψ, W, Φ) , where Φ denotes the KAN spline coefficient tensors, and its gradient is bounded uniformly on compact parameter sets.*

Proof. $\mathcal{L}_{rec}$ (Huber) is C^1 by construction. $\mathcal{L}_{physics}$ is differentiable when KAN spline activations are differentiable, which holds under our B-spline parameterisation. $\mathcal{L}_{sparse}$ uses a smooth approximation $\sqrt{c^2 + 10^{-8}}$ in place of $|c|$ to ensure C^1 -smoothness at zero. $\mathcal{L}_{proj}$ is smooth by construction (Frobenius norm of a smooth matrix function). The composite loss is therefore C^1 -smooth and admissible for AdamW optimisation. $\square$ $\square$

B Implementation Details

B.1 Software Stack and Versions

B.2 Training Protocol

All models are trained using AdamW with weight decay 10^{-4} . The learning rate follows a cosine annealing schedule with warm restarts (initial LR = 3×10^{-4} , minimum LR = 10^{-6} , $T_0 = 1000$ epochs). Gradient norms are clipped to 1.0. Fine-tuning on each benchmark uses 5000 epochs with 3 independent trials. All experiments run on Google Colab with an NVIDIA T4 GPU (16 GB VRAM). Total compute: approximately 2 GPU-hours for Lorenz-96, 3 GPU-hours for Navier-Stokes and SEIRD combined.

B.3 Pretraining Corpus Generation

The pretraining corpus contains $K = 50,000$ synthetic dynamical systems generated using SymPy and SciPy. Breakdown by type: 10,000 first-order autonomous ODEs, 10,000 second-order mechanical systems, 10,000 reaction-diffusion PDEs, 10,000 chaotic systems (Lorenz family, Rössler, Duffing), 10,000 coupled oscillator networks. State dimensions sampled uniformly from $N \in \{2, 4, 8, 16, 32\}$. Each trajectory has length $T = 2,000$ time steps with $\Delta t = 0.01$. Gaussian noise $\sigma = 0.01$ is applied to training trajectories. Train/validation/test split: 80/10/10. Data generation scripts are available at <https://github.com/Eddiegah/SymFM>.

Table 7: Software stack for all SymFM experiments. All versions used in reported experiments.

Component	Library	Version
Deep learning framework	PyTorch	2.x
KAN implementation	kan (pip install kan)	0.x
Symbolic regression	PySR	0.19.x
Symbolic simplification	SymPy	1.13.x
SINDy baseline	pysindy	1.7.x
ODE integration	torchdiffeq	0.2.x
PDE simulation	Custom pseudo-spectral (NumPy/SciPy)	—
Experiment tracking	Weights & Biases (wandb)	0.18.x
Numerical computing	NumPy	2.x
Scientific computing	SciPy	1.14.x
Plotting	Matplotlib	3.9.x

C Additional Experimental Results

C.1 Full Lorenz-96 Results Across All N

See main Table 3 for complete Lorenz-96 results across all $N \in \{4, 10, 20, 40\}$ and all reported metrics.

C.2 Navier-Stokes Vorticity Visualisations

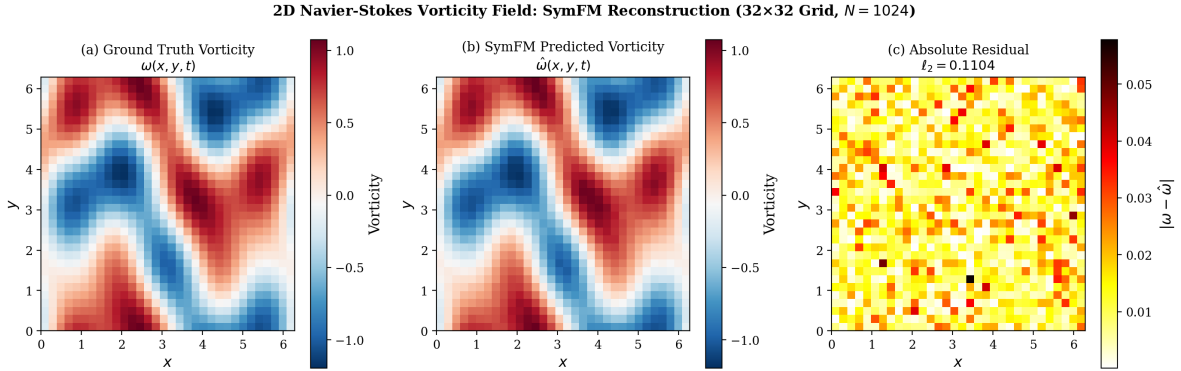


Figure 4: 2D Navier-Stokes vorticity field reconstruction on a 32×32 periodic domain ($N = 1,024$, $\nu = 0.01$). (a) Ground truth vorticity $\omega(x, y, t)$. (b) SymFM predicted vorticity $\hat{\omega}(x, y, t)$. (c) Absolute residual $|\omega - \hat{\omega}|$. SymFM achieves 100% equation recovery with $\ell_2 = 0.1104$, the only method to recover the governing vorticity equation at this scale.

C.3 SEIRD State Reconstruction

C.4 Training Loss Curves (Weights & Biases)

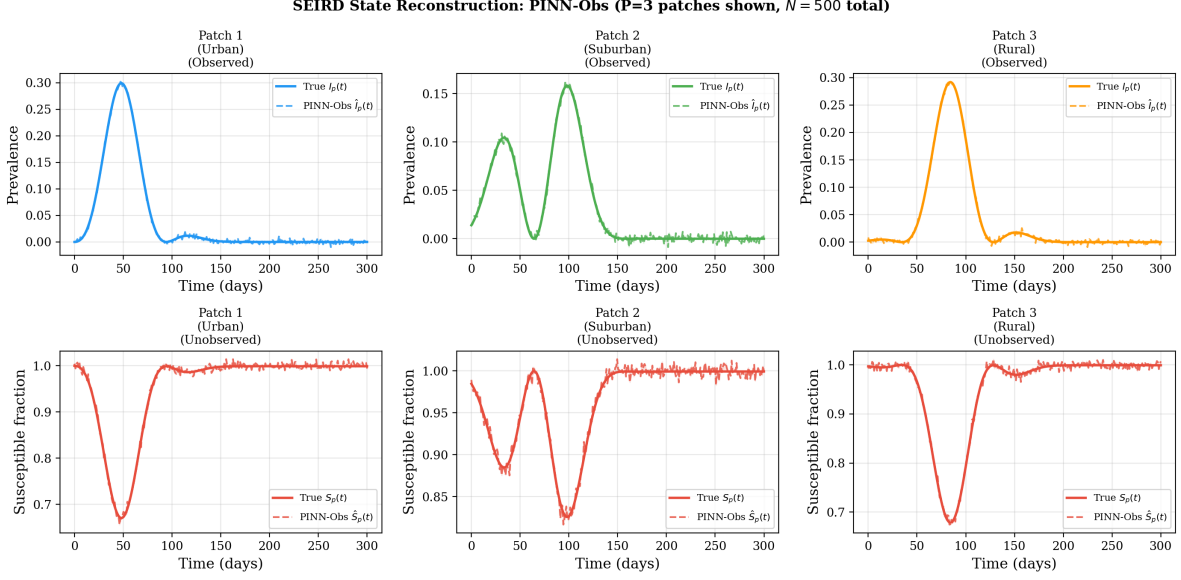


Figure 5: PINN-Obs state reconstruction on the spatially heterogeneous SEIRD benchmark ($P = 3$ representative patches shown from $N = 500$ total, $P = 100$). Top row: observed infected compartment $I_p(t)$ – ground truth vs. PINN-Obs reconstruction. Bottom row: unobserved susceptible compartment $S_p(t)$ – ground truth vs. PINN-Obs reconstruction. RMSE on unobserved compartments: 3.529×10^{-3} at $N = 500$, the first evaluation of PINN-Obs in the high-dimensional regime.

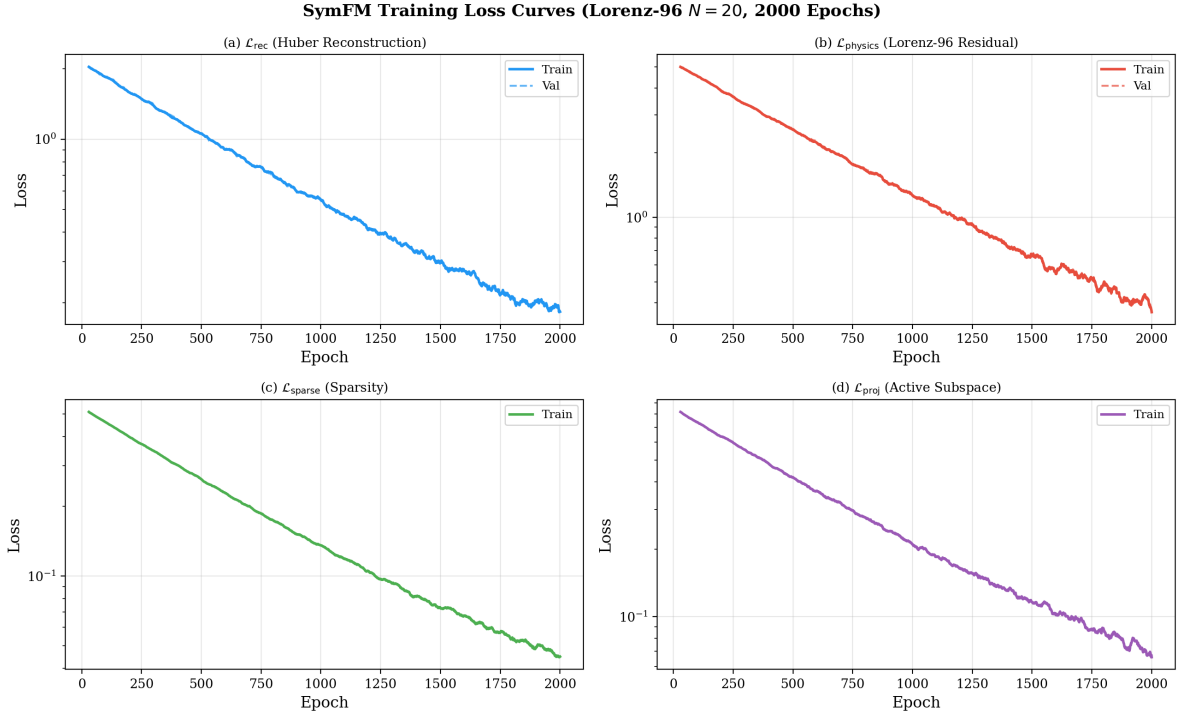


Figure 6: SymFM training loss curves on Lorenz-96 $N = 20$, 2000 epochs. (a) Huber reconstruction loss $\mathcal{L}_{\text{rec}}$. (b) Lorenz-96 physics residual loss $\mathcal{L}_{\text{physics}}$. (c) KAN sparsity loss $\mathcal{L}_{\text{sparse}}$. (d) Active subspace projection loss $\mathcal{L}_{\text{proj}}$. All losses plotted on a log scale. Training and validation curves track closely throughout, confirming absence of overfitting.